CASE STUDY ON AI-BASED CRICKET HIGHLIGHT GENERATION SYSTEM USING C++ AND MULTI-MODAL EVENT DETECTION

Problem Statement:

Cricket matches, especially ODIs and Test matches, are long in duration, making it difficult for users to watch the entire match. Existing highlight generation is mostly manual, time-consuming, and may miss important moments.

There is a need for an automated system that can efficiently generate a concise 10-minute highlight video from full-length cricket matches.

Objective:

- Automatically detect important moments in a cricket match
- Identify peak events using crowd reaction and gameplay signals
- Extract and rank key events such as wickets and boundaries
- Generate a 10-minute highlight video
- Reduce manual effort and enable faster highlight creation

Proposed Idea:

The proposed system is a **C++-based intelligent video processing pipeline** that automatically generates highlights by analyzing both audio and visual cues.

Key Steps:

1. Input full cricket match video
2. Extract audio and video frames
3. Detect important events using multiple signals
4. Score and rank detected events
5. Extract short clips around high-score events
6. Merge clips into a 10-minute highlight video

METHODOLOGY

The system uses a multi-modal approach combining different techniques:

Audio Analysis:

The system analyzes the audio component of the cricket match to identify moments of high excitement. During important events such as wickets, boundaries, or milestones, crowd noise and commentary intensity typically increase significantly. The audio is first extracted from the input video using FFmpeg and processed to detect peaks in amplitude. These peaks are treated as indicators of potential highlight moments. By setting appropriate thresholds, the system distinguishes between normal background noise and significant crowd reactions, thereby identifying segments that are likely to be important.

Motion Detection:

Motion detection is used to identify visually dynamic scenes in the match. Important cricket events such as fast deliveries, boundary shots, or player celebrations involve rapid changes between consecutive frames. Using OpenCV, the system converts frames to grayscale and computes the difference between successive frames. A higher difference indicates increased motion, which may correspond to a highlight-worthy event. This method helps in capturing gameplay intensity and complements audio-based detection by providing visual confirmation of action.

Face Detection (Crowd Excitement):

To further enhance highlight detection, the system incorporates face detection to analyze crowd reactions. Using Haar Cascade classifiers in OpenCV, the system detects and counts the number of faces present in each frame. During exciting moments, camera shots often shift to the audience, resulting in a higher number of detected faces. Additionally, a sudden increase in face count between frames may indicate crowd excitement. This feature provides an indirect but effective measure of audience engagement, which is a key indicator of significant match events.

Scoreboard Change Detection:

Scoreboard analysis is used to detect actual game events such as runs scored or wickets taken. A specific region of interest (ROI) corresponding to the scoreboard is defined in the video frame. The system compares this region across consecutive frames to identify pixel-level changes. Significant differences indicate updates in the score or match status. Since scoreboard changes directly reflect key events in the game, this method provides a reliable way to detect important moments and reduces dependence on indirect indicators like motion or audio alone.

Event Scoring and Ranking:

All detected features—audio peaks, motion intensity, face count, and scoreboard changes—are combined to compute a final score for each event. Each feature contributes a weighted value to the overall score, ensuring a balanced evaluation of importance. Events with higher scores are considered more significant. The system then ranks these events in descending order and selects the top events that collectively fit within a 10-minute duration. This scoring mechanism ensures that the most impactful moments are included in the final highlight video.

System Architecture:

```
Input Video
  ↓
Audio Extraction + Frame Processing
  ↓
Feature Detection:
  - Audio Peaks
  - Motion Detection
  - Face Detection
  - Scoreboard Change
  ↓
Event Scoring & Ranking
  ↓
Clip Extraction (FFmpeg)
  ↓
Highlight Generation (10 mins)
```

Technologies Used:

- **C++** – Core implementation
- **OpenCV** – Video processing and face detection
- **FFmpeg** – Audio extraction and video clipping
- **Haar Cascade Classifier** – Face detection

Expected Output:

Automatically generated **10-minute highlight video** that includes:

- ✓ Wickets
- ✓ Boundaries
- ✓ Crowd reactions
- ✓ Important match moments

Advantages:

- Eliminates manual video editing
- Faster highlight generation
- Scalable for multiple matches
- Real-time processing potential

Challenges:

- Accurate detection of events in varying conditions
- Noise in audio signals
- Scoreboard position variability
- Performance optimization

Conclusion

This system demonstrates an efficient and scalable approach to automatic cricket highlight generation using C++. By combining audio, motion, face detection, and scoreboard analysis, the system effectively identifies key moments and produces concise highlight videos.